

PURPOSE

Sepsis is the leading cause of in-hospital mortality in the United States, responsible for 1 in 5 deaths globally (Singer et al., 2016). In the US alone, sepsis accounts for **\$62 billion** in annual healthcare expenditures, more than any other disease. Despite this burden, there are **no FDA-approved drugs** that specifically target the underlying molecular driver of sepsis: dysregulated protease cascades.

During the onset of sepsis, infection triggers an overwhelming immune response that activates dozens of proteases simultaneously. These proteases normally serve critical regulatory functions in immunity, coagulation, and wound healing. In sepsis, however, their coordinated dysregulation drives a self-amplifying loop causing **irreversible multi-organ failure** (Gando et al., 2016).



Figure 1: Sepsis large-scale impact on United States residents in 2025 the Assembly, 2025).

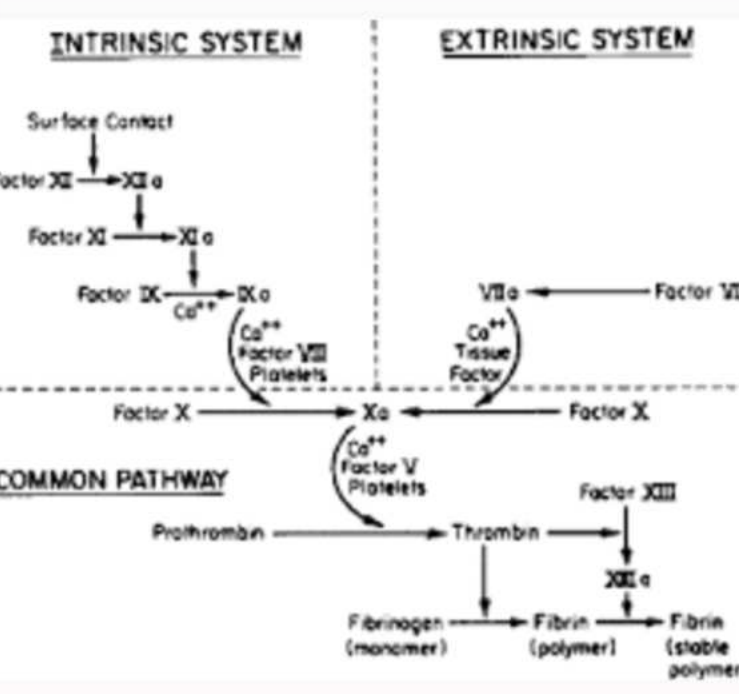


Figure 2: Coagulation Cascade (Science Direct, 2020)

Current solutions for Inhibitor Discovery Are Insufficient:

Static predictions and Low Accuracy

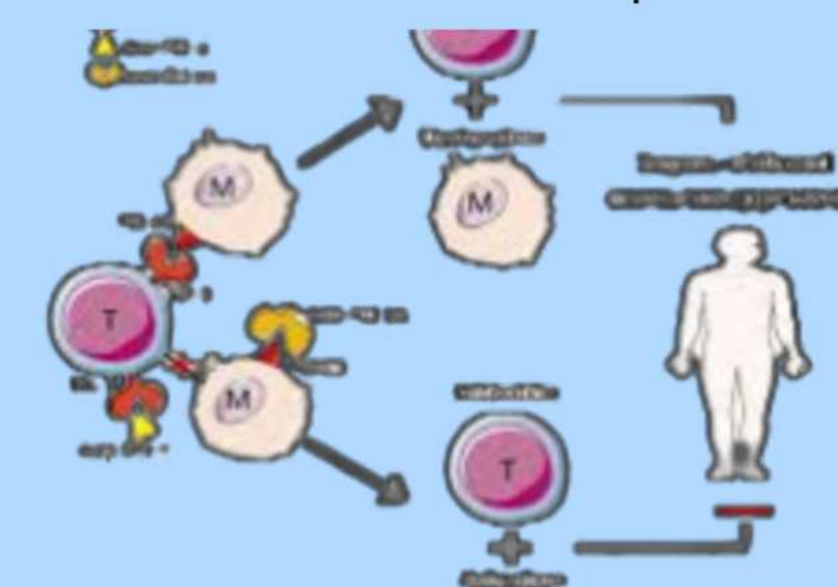
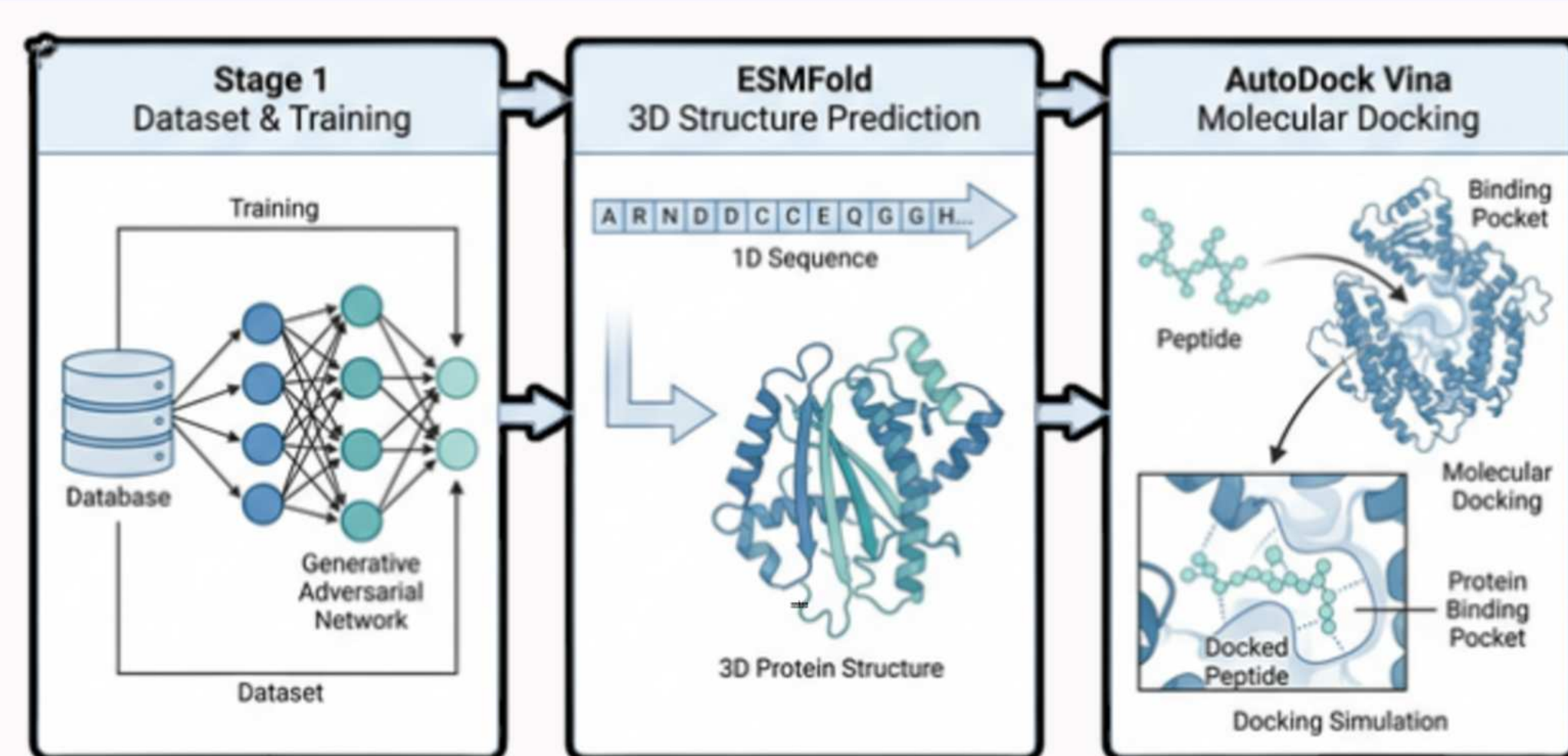


Figure 3: Current Detection method and treatment of Sepsis (MDPI, 2019)



Figure 4: Length, Cost, and Inaccuracies of current Discovery. (Ze Clinics, 2025)

Solution Overview



(Image Created by Researcher)

Dataset

Data Collection

- 11,551 experimentally confirmed proteolytic cleavage events extracted from MEROPS database (Release 12.4)
- High-resolution 3D crystal structures for 27 target proteases sourced from the RCSB Protein Data Bank (PDB)

Four Pathological Protease Classes

- Matrix Metalloproteinases (MMPs): Tissue-destructive
- Caspases: Drivers of cellular apoptosis
- Serine Proteases: Coagulation and inflammatory cascades
- Cysteine Proteases: Cytotoxic pathways

Figure 4: Step by step diagram of the Data Preprocessing that occurred prior to training of the pipeline (Image Created by Researcher Using Canva).

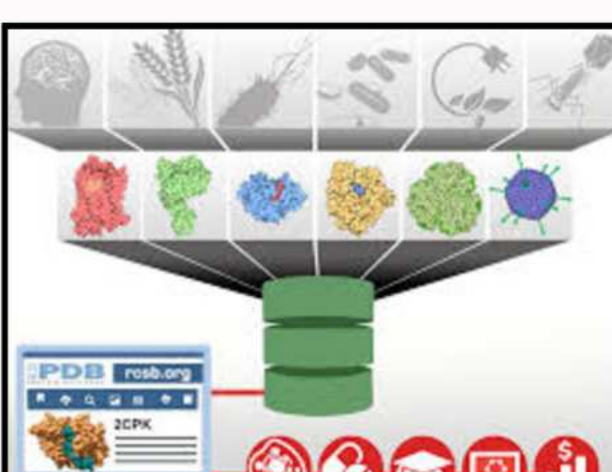
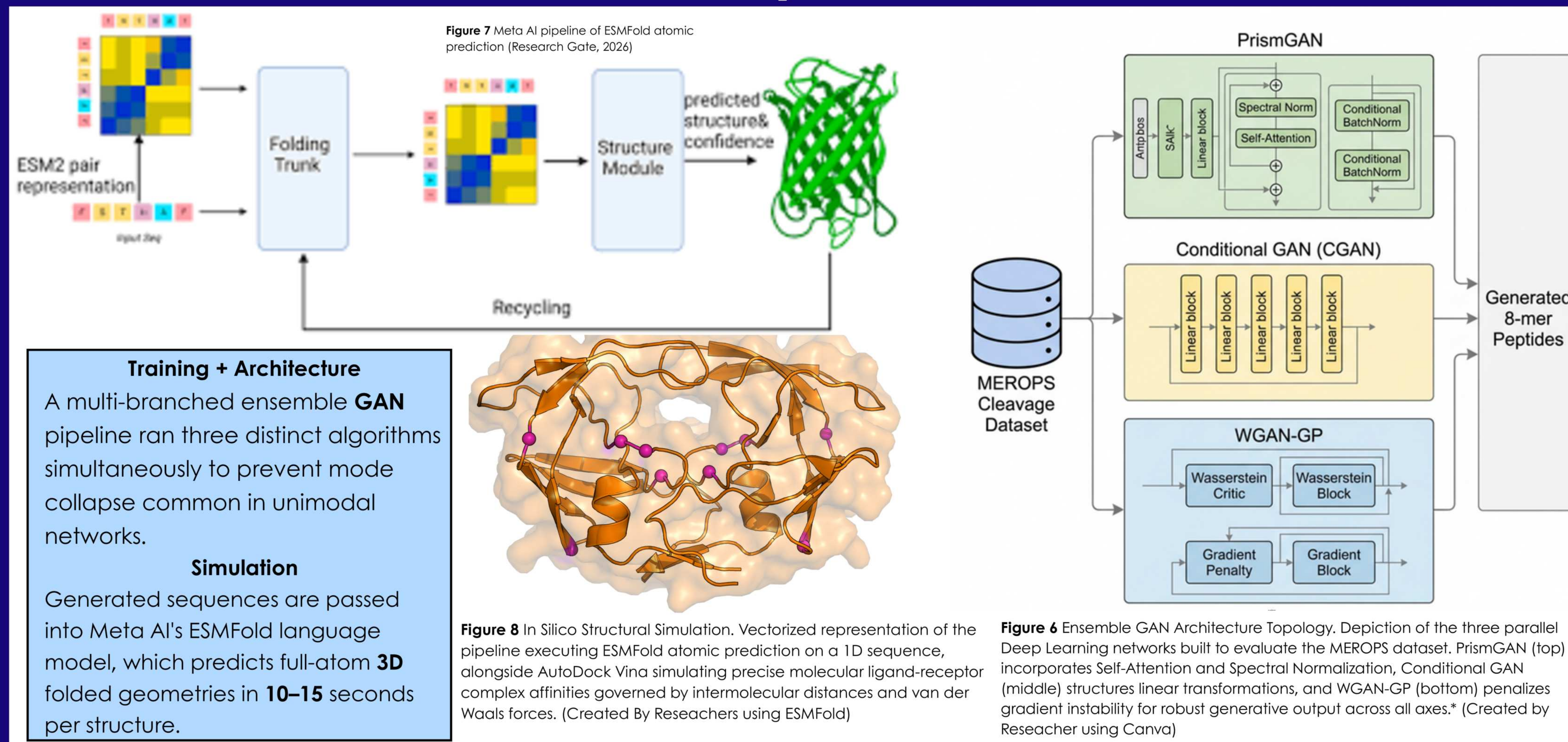


Figure 5: The RCBS Protein Data Bank

SPECTRA: De Novo Broad-Spectrum Sepsis Inhibitor Design Across 27 Protease Targets via a Multi-Architecture Generative Adversarial Pipeline



Training + Architecture

A multi-branched ensemble GAN pipeline ran three distinct algorithms simultaneously to prevent mode collapse common in unimodal networks.

Simulation

Generated sequences are passed into Meta AI's ESMFold language model, which predicts full-atom 3D folded geometries in 10-15 seconds per structure.

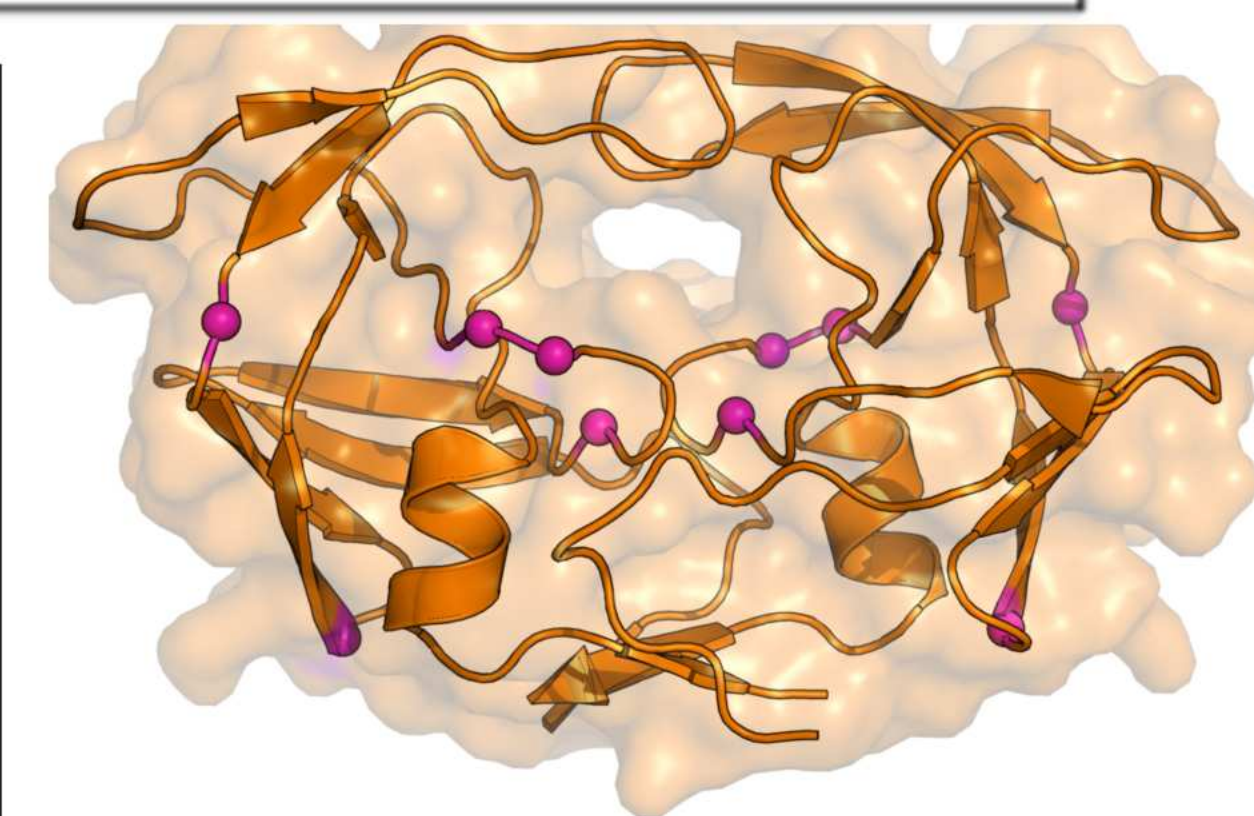


Figure 8: In Silico Structural Simulation. Vectorized representation of the pipeline executing ESMFold atomic prediction on a 1D sequence, alongside AutoDock Vina simulating precise molecular ligand-receptor complex affinities governed by intermolecular distances and van der Waals forces. (Created By Researchers using ESMFold)

Figure 6: Ensemble GAN Architecture Topology. Depiction of the three parallel Deep Learning networks built to evaluate the MEROPS dataset. PrismGAN (top) incorporates Self-Attention and Spectral Normalization, Conditional GAN (middle) structures linear transformations, and WGAN-GP (bottom) penalizes gradient instability for robust generative output across all axes.* (Created by Researcher using Canva)

Model Performance



Figure 9: Per-Protease Head-to-Head Algorithm vs. Nature Performance. Created By Researchers using Matplotlib

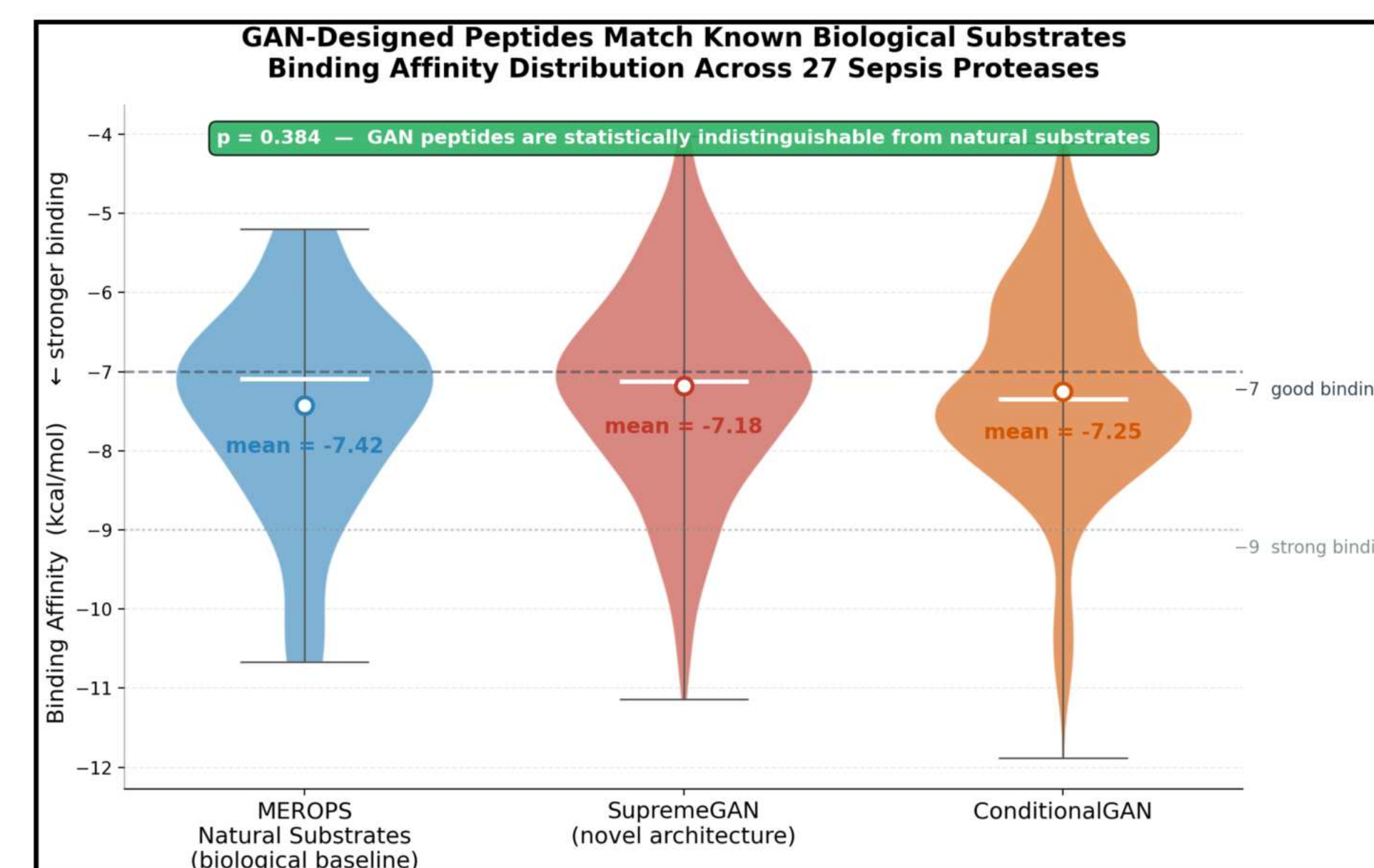


Figure 10: Binding Affinity Distribution & Equivalence (Comparison between baseline and GAN's). (Created By Researchers using Matplotlib)

Inhibitor Optimization & ADMET Profiling



Figure 11: Inhibitor Optimization Structure and Process (Created by researcher using Canva)

Highly-rated **reversible** peptide substrates were computationally upgraded into irreversible **covalent inhibitors** by attaching a chemical **warhead** designed to permanently block the enzyme's active site.



Figure 12: ADMET Profiling factors (Created by Reachers using Canva)

These optimized covalent inhibitors underwent comprehensive **computational profiling** to ensure highly favorable Absorption, Distribution, Metabolism, Excretion, and Toxicity limits for safe human administration during systemic sepsis.

Discussion

Performance Analysis

The procedurally AI-generated candidate panel culminated into **28** total verified "excellent binders". Panels binding affinity spans **-4.93** to a highly potent **-11.88 kcal/mol**.

Statistical Testing

A two One-Sided Test algorithm (TOST) was used with a severity margin of $\delta = \pm 0.75$ kcal/mol (Trott & Olson, 2010). The pipeline proved to be **equivalent** MEROPS baselines (bounds perfectly measured at statistically significant limits of $p_{lower} < 0.001$, $p_{upper} = 0.011$).

Future Development

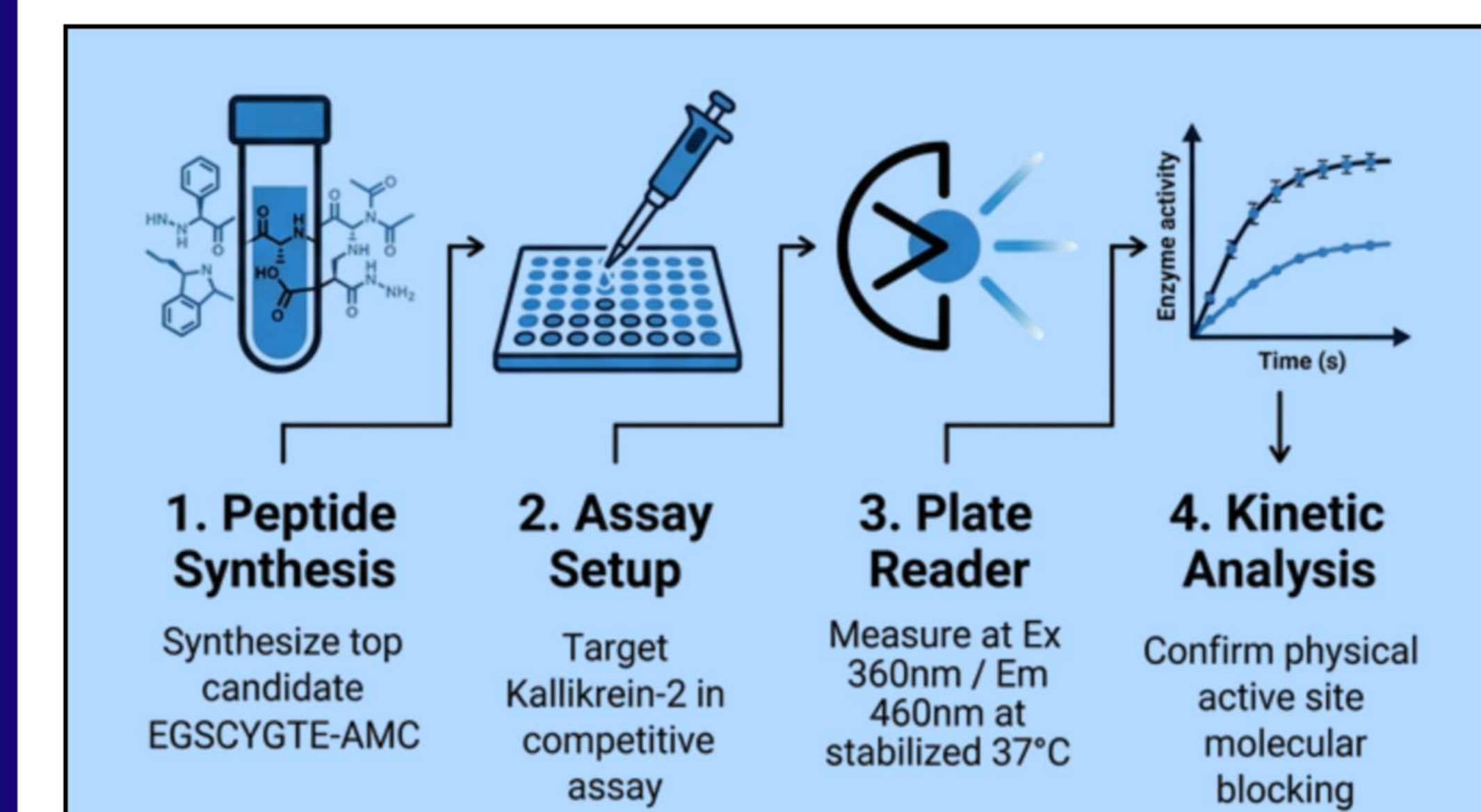


Fig 11. Image Created by Researcher using Canva

Empirical Wet-Lab Validation:

- 9.72 kcal/mol *EGSCYGTG-AMC* top pipeline candidate.

Molecular Dynamics Trajectory:

- Simulating peptide within fully explicit physical solvent **simulations** to detail precise **MM-GBSA** kinetic stability (Fig 12)

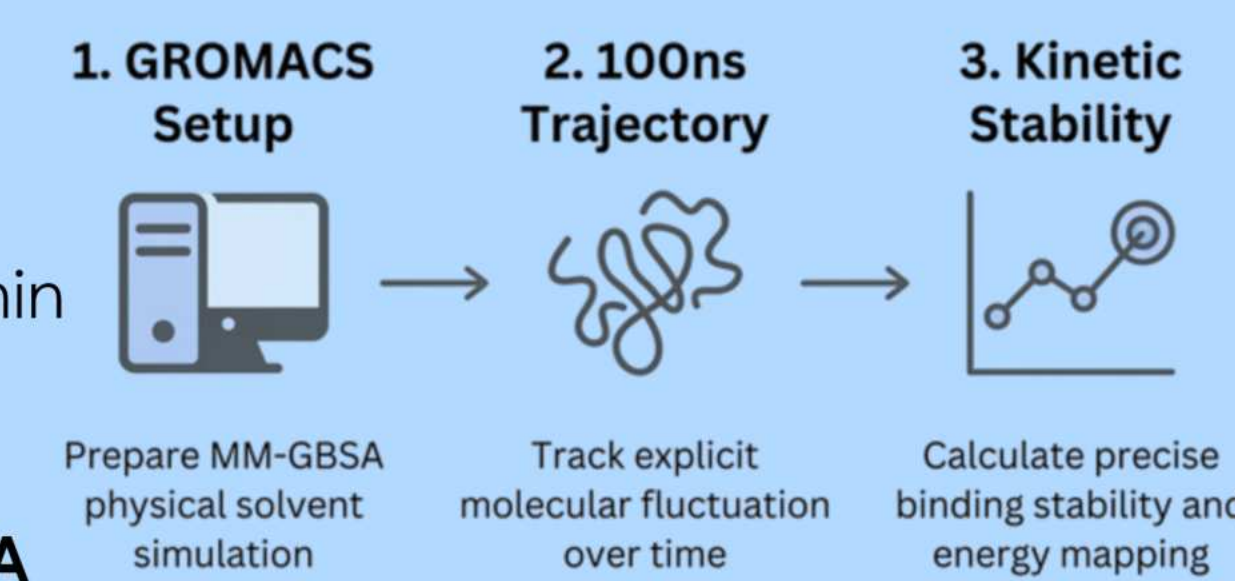


Fig 12. Image Created by Researcher using Canva

Conclusion

AI Outperforms Nature: Generative models exceeded mammalian evolution in 25 out of 27 targets.

Multi-Target Efficacy: The single pipeline effectively neutralizes dozens of pathways at once.

Universal Architecture: Framework seamlessly redirects to other multi-pathway diseases.

Fig 13. Image Created by Researcher using Canva

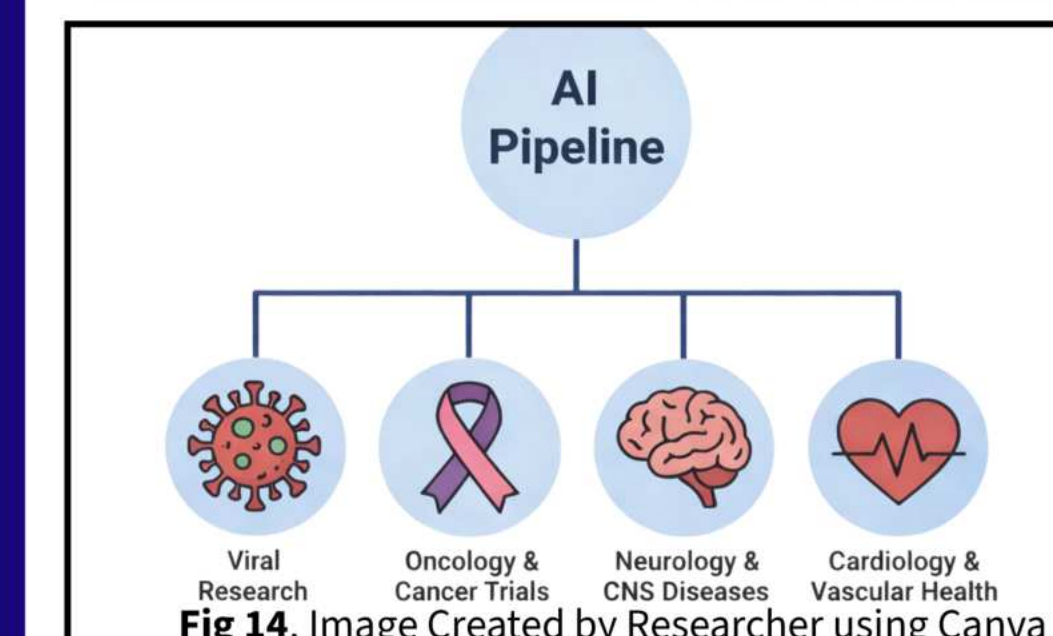


Fig 14. Image Created by Researcher using Canva

The multi-architecture GAN framework is universally scalable, easily translating from Sepsis to combat Cancers and Alzheimer Pathology (Fig 14)